Advanced JavaScript

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iTi

Day3

Property Descriptors

Property Descriptors

- Property descriptors hold descriptive information about object properties.
- Property Descriptors allows developer to control some of the internal attributes of the object properties it can be either
  - Data Descriptor or,
  - Accessor Descriptor
- To define Property Descriptors use
  Object.defineProperty(obj,”prop”,{})
  Object.defineProperties(obj,{})

Data Descriptor

- A *data descriptor* is a property that has a value, which may be read-only. It is represented by the following keys
  - value : the value associated with the property. Default value is undefined.
  - writable : a Boolean value that determines whether or not the property value can be changed within an assignment operator.
    Default value is false.

Accessor Descriptor

- An *accessor descriptor* is a property described by a getter-setter pair of functions. It is represented by the following keys.
  - get : A function which serves as a getter for the property.
    Default is undefined.
  - set : A function which serves as a setter for the property.
    Default is undefined.

Data & Accessor Descriptors Shared Fields

- Both data and accessor descriptors are objects. They share the following optional keys:
  - configurable : determines whether or not a property descriptor can be changed, and the property can be deleted. Default is false.
  - enumerable : determines whether or not the property is enumerated with all of the other members. Default is false.
    i.e. the property will be iterated over when a user does for (var prop in obj){} (or similar).

Descriptors Identifying Fields

Fields	DATA DESCRIPTOR	ACCESSOR DESCRIPTOR	Default Value
value	✓		undefined
writable	✓		false
enumerable	✓	✓	false
configurable	✓	✓	false
get		✓	undefined
set		✓	undefined

Data Descriptors Example

```
var Employee = function(nme, age){
    var person = {};

    Object.defineProperty (person, “nm”, {value : nme, writable :
       true, configurable: true, enumerable: true } );

    Object.defineProperty (person, “age”, {value : age} );

    Object.defineProperty (person, “show”, {value : function (){
                alert("Employee " + this.nm + " is " + this.age
                     + " years old.");
        }
    } );

    return person;

    }
```

Data Descriptors Example

```
var Employee = function(nm, age){
        var person = {};

        Object.defineProperties (person,{
                nm:{
                        value : nm,
                        writable : false},
                age:{…..},
                show:{…..}
        ……..

        } );

        return person;

    }
```

Accessor Descriptors Example

```
var Employee = function(name, age){
var emp= {};
Object.defineProperty (emp, “nm", {
    get : function() { return name; },
    set : function(val) { name = val; }
    });
return emp;
}
var e= new Employee();
e.nm = “Nour";
var t_emp = e.nm; // alert(t_emp)
```

Example!

value, get & set fields

- An object property cannot have both the *value* and *getter/setter* descriptors. You've got to choose one.
- *Value* can be pretty much anything
  - i.e. primitives or built-in types or even be a function.
- You can use the *getter* and *setters* to mock read-only properties.
- You can even have the *setter* throw Exceptions when users try to set it.

Reminder : Object Object Properties & Methods

- .hasOwnProperty("prop")
- .valueOf()
- .toString()

- Object.keys(obj) → enumerable properties
- Object.getOwnPropertyNames(obj) → enumerable and non-enumerable properties

- Object.defineProperty(obj,"prop",{})
- Object.defineProperties(obj,{})

- Object.getOwnPropertyDescriptor(obj,prop)
- Object.getOwnPropertyDescriptors(ctor.prototype)

- Object.create(obj [,{}])
- …

Other Useful Object Methods

- Object.seal()
  - Marks every existing property on the object as *non-configurable*
  - Then call *Object.preventExtensions* to prevent adding new properties

- Object.freeze()
  - Mark every existing property on the object as non-writable
  - Invokes *Object.seal* to prevent adding new properties and marks existing properties as non-configurable

in Operator
vs
.hasOwnProperty()

Prototype Property

Prototype Property

- **Prototype:** is a property that allows you to add more properties and methods to any created object.
- It is a property of the function objects that gets created as soon as you define a function.
- Attaching new properties to a prototype will make them a part of every object instantiated from the original prototype, effectively making all the properties public (and accessible by all).
- This is another way to add more functionality to already created objects using constructor function
- It is also used for inheritance

Prototype Property & Public Method

- Public methods are completely accessible by the end user.
- Public method is a property of the function objects
- To achieve these public methods, which are available on every instance of a particular object, we need to the *prototype* property

Prototype Property & Public Method

```
function User( name, age ){
    this.name = name;
    this.age = age;
}
```

```
var User = function (name,age){
    this.name = name;
    this.age = age;
}
```

```
// Add a public accessory method for name
User.prototype.getName = function(){
return this.name;};

// Add a public accessory method for age
User.prototype.getAge = function(){
return this.age; };

User.prototype.job="Engineer";
```

Pseudo classical pattern

Prototype Property & Public Method

```
// Instantiate a new User object

var user = new User( “Ahmed", 25 );

alert( user.getName()); //Ahmed

alert( user.getAge()); //25

alert(user.job); //Engineer
```

Example!

Overriding

occurs when two methods having the same method name and parameters (i.e., *method signature*) where one of the methods is implemented in the parent class while the other is implemented in the child class, so that a child class provides a specific implementation of a method that is already provided its parent class.

Prototype Property & Overriding Methods

- override methods when its required to be different from the available property

```
// overridding toString() for User object
User.prototype.toString = function(){
return "user name is: "+this.name+"and his age is: "+this.age;};

document.write(user.toString().);
// user name is: Ahmed and his age is: 25
```

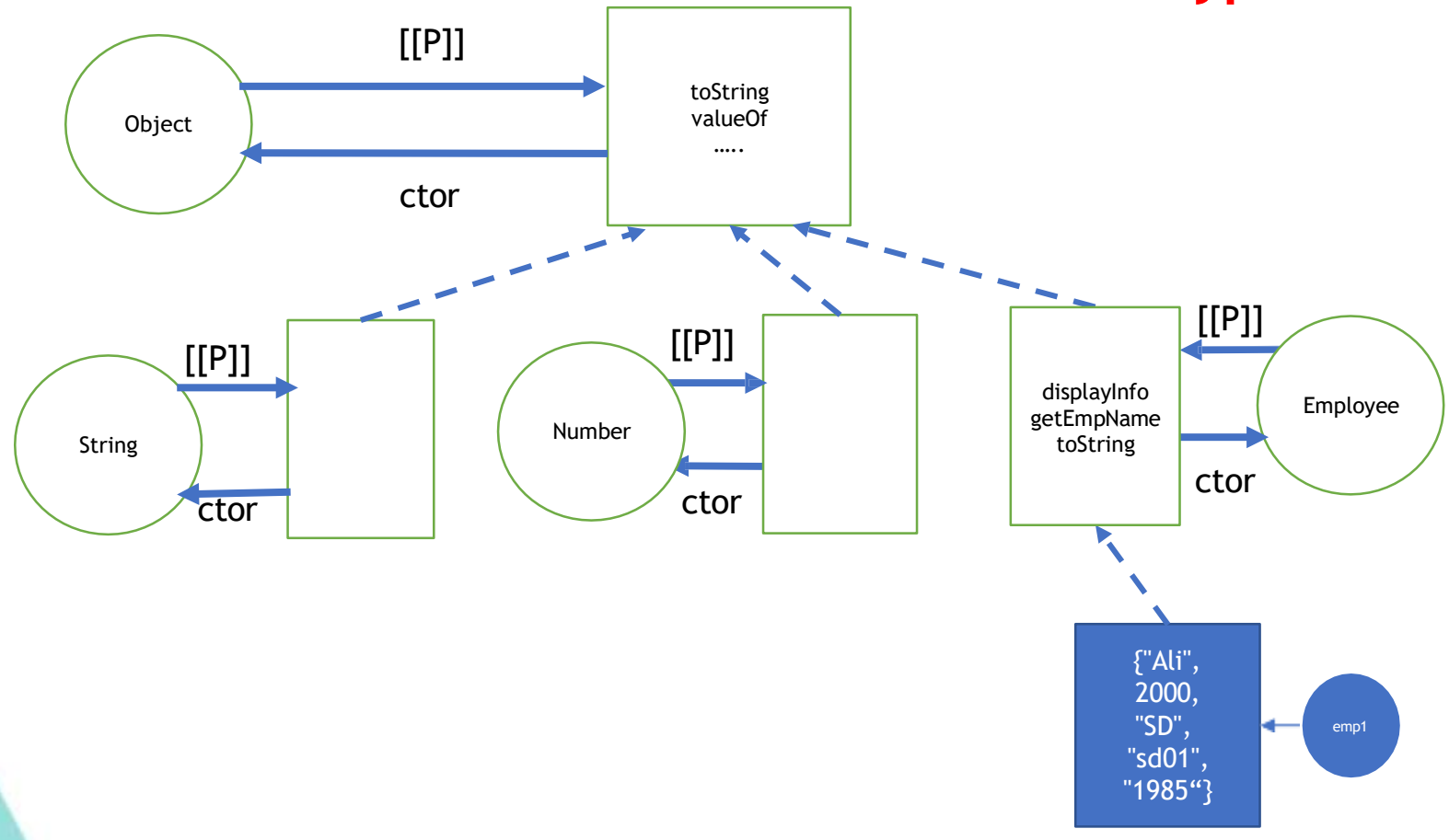
Prototype Chaining

- Prototype chaining is used to build new types of objects based on existing ones. It has a very similar job to inheritance in a class based language

- A mechanism for making objects that resemble another object when we want these object to have same properties

- Make one object behave as if it has all of the properties of another object by delegating a lookups from the 1st to the 2nd

Prototype Chaining



var emp1= new Employee(“ali”,200,”SD”,”sd01”,”1985”)

Inheritance

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JavaScript is a prototype based language

– Fourth level

» Fifth leve

It is also known as instance-based programming, prototype-oriented, prototypal, or classless.

https://en.wikipedia.org/wiki/Prototype-based_programming

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prototype based language is a style of object-oriented programming in which behavior reuse (known as inheritance) is performed via a process of reusing existing objects that serve as prototypes

https://en.wikipedia.org/wiki/Prototype-based_programming

Inheritance

- Inheritance refers to an object being able to inherit methods and properties from a parent object (a Class in other OOP languages, or a Function in JavaScript).

- There are different ways for a child to inherit from a parent
  - Parasitic Inheritance
  - Prototypal Inheritance
  - etc..

Inheritance in JavaScript works through something called prototypes and this form of inheritance is often called prototypal inheritance

Prototypal Inheritance

- JavaScript uses a unique form of object creation and inheritance called *prototypal inheritance*

- It is the default way to implement inheritance, and is described in the ECMAScript standard.

- The premise behind this is that an object constructor can inherit methods from one other object, creating a *prototype object from which all other new objects are built.*

Prototypal Inheritance

- This process is facilitated by the *prototype property.*
- Prototypes do not inherit their properties from other prototypes or other constructors; they inherit them from physical objects.
- Prototype chaining (pseudo-classical)

```
Child.prototype = new Parent();
```

- Inherit only the prototype

```
Child.prototype = Parent.prototype;
```

Prototypal Inheritance

```
// Using prototype object for inheritance behavior
        function Employee(name, age){
            this.name = name;
            this.age = age;
        }

// No Classes, use constructor functions
        function Person(name){
            this.name = name;
        }
// Public functions using prototype
        Person.prototype.getName = function() {
            return this.name;
        }
```

Prototypal Inheritance

```
// Inherit only the prototype
    Employee.prototype = Person.prototype;

   var e = new Employee('Ahmed', 12);
   e.getName();    //Inherited from Person
```

```
// prototype chaining
    Employee.prototype = new Person();

   var e = new Employee('Ahmed', 12);
   e.getName();    //Inherited from Person
```

Parasitic Inheritance

- Parasitic inheritance was first proposed as a way of handling inheritance in JavaScript by Douglas Crockford in 2005.

- In this methodology you create an object by using a function that copies another object, augments it with the additional properties and methods you want the new object to have and then returns the new object.

```
var childObjProto=Object.create (parentObjProto)
```

Parasitic Inheritance

- using create() creates an empty object , its proto is pointing to the object passed as a parameter

```
// prototype chaining
    Employee.prototype = Object.create(Person.prototype);

   var e = new Employee('Ahmed', 12);
   e.getName();    //Inherited from Person
```

Parasitic inheritance
is said to be
Pure Prototypal Inheritance.

In
Prototype Chain Mechanism
if an object does not know how
to retrieve a property,
it tries to ask the object
above in the chain.
It **delegates**.

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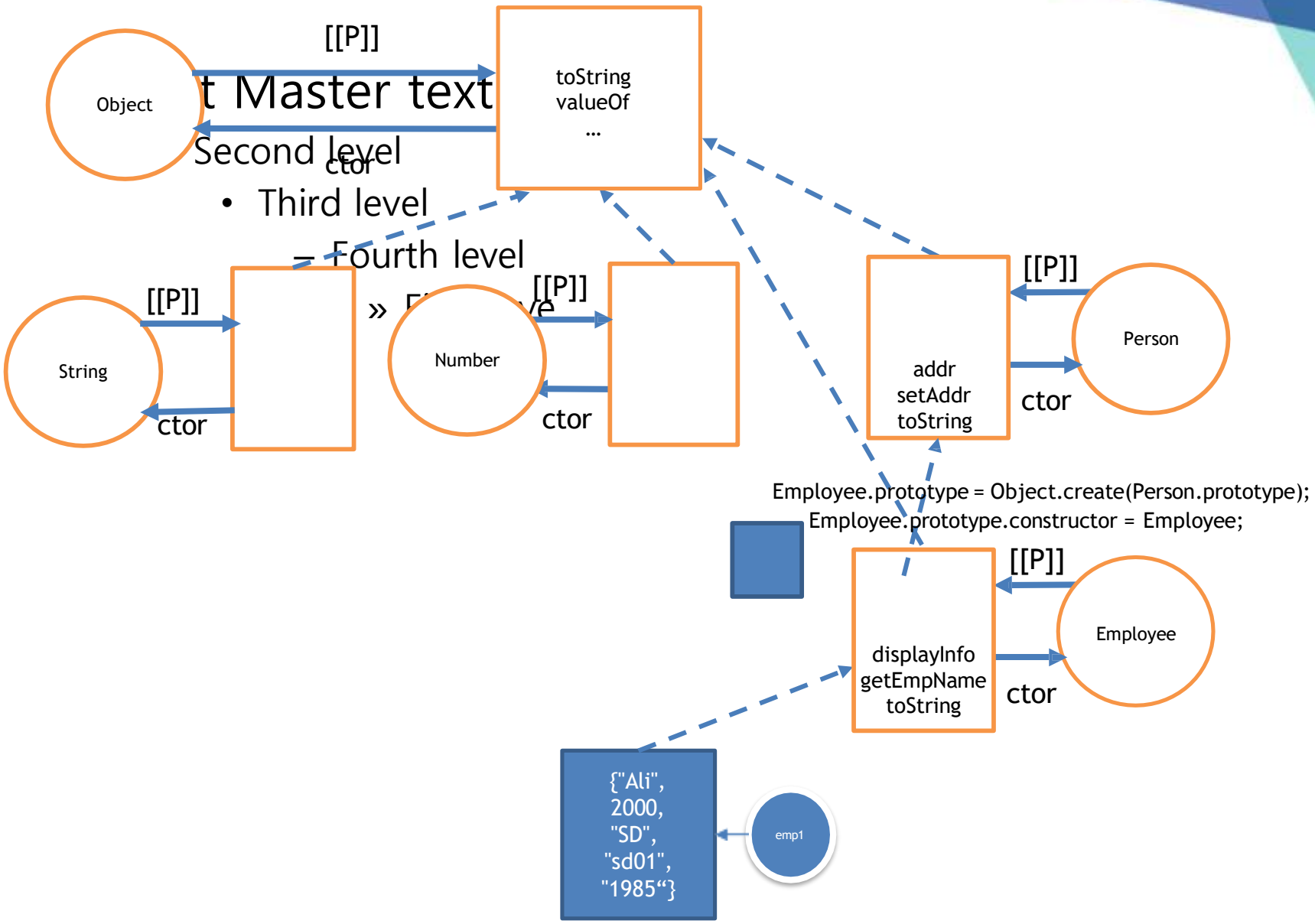

var emp1= new Employee("ali",200,"SD","sd01","1985")

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Second level

- Third level
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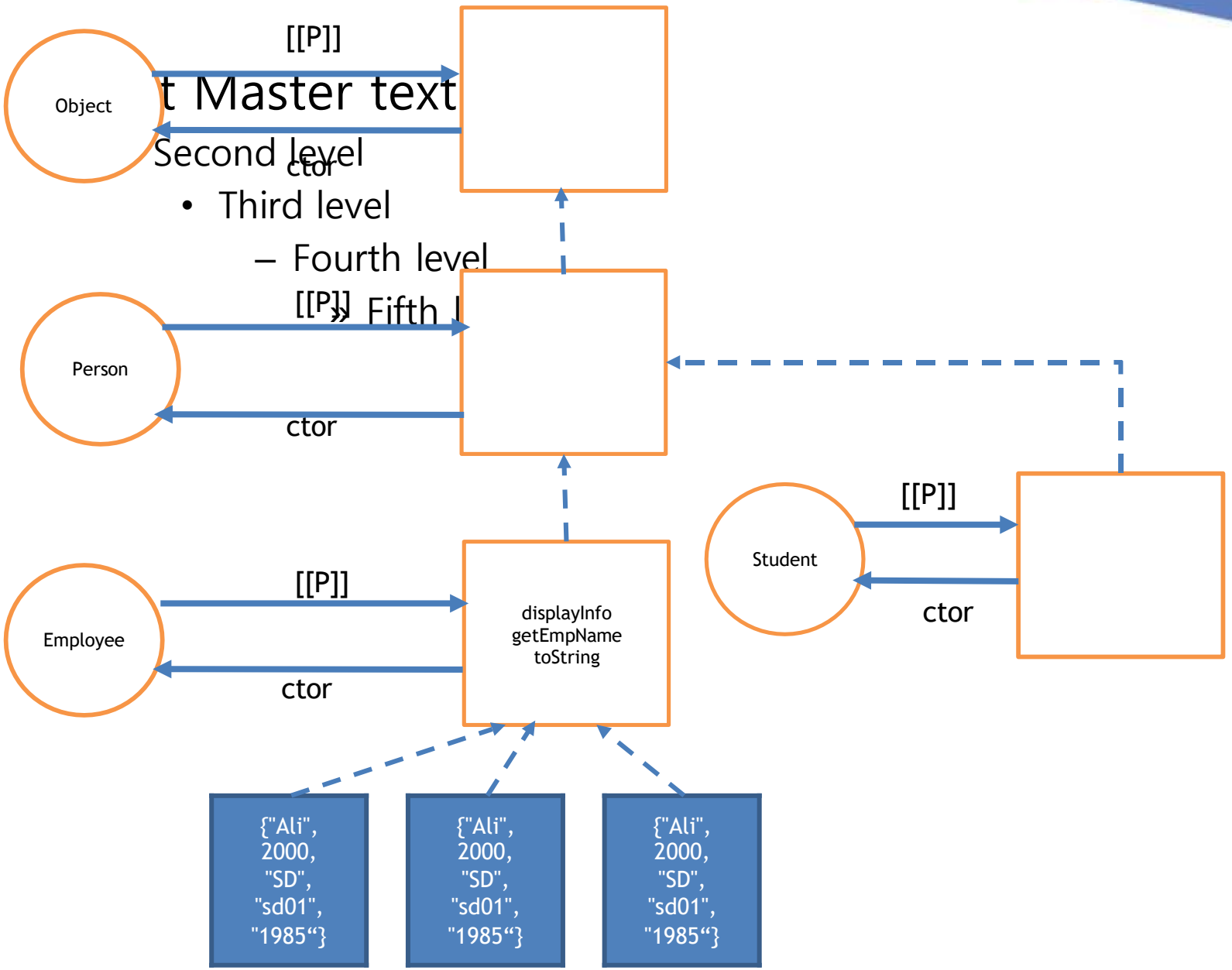

Change your thinking from the class/inheritance design pattern to the behavior delegation design pattern.

Reminder

- Object & Object creation
  - Factory
  - Constructor
  - Literal Pattern
- this and new operator
- Class property and method
- Scope & Closure
  - Local scope variable = Private variable
  - Inner function = Private methods
- Privileged methods = getters & setters
- Property descriptor
  - Data descriptor
  - Accessor descriptor
- Prototype property
- Inheritance & delegation behavior

By the end of development phase

- Validating HTML
  - https://validator.w3.org/#validate_by_input
- Validating CSS
  - https://jigsaw.w3.org/css-validator/#validate_by_input
- Minifing & Compression
  - https://jscompress.com/
  - https://javascript-minifier.com/
  - https://www.minifier.org/
  - https://www.danstools.com/javascript-obfuscate/

Assignment

Assignment